Class 10 Notes

Quick revision: Maths, English, Science

Maths

- **Real numbers:** Euclid's division lemma: $a = bq + r$, $0 \le r < b$. Fundamental theorem of arithmetic: every composite number has a unique prime factorisation. HCF × LCM = product of the two numbers.
- **Polynomials:** For $ax^2 + bx + c$: sum of zeroes = $-b/a$, product of zeroes = c/a. Zeroes are where the graph cuts the x-axis.
- **Pair of linear equations:** $a_1/a_2 \neq b_1/b_2$: unique solution. $a_1/a_2 = b_1/b_2 \neq c_1/c_2$: no solution. All three equal: infinitely many solutions.
- **Quadratic equations:** $ax^2 + bx + c = 0$. $x = [-b \pm \sqrt{(b^2 - 4ac)}] / 2a$. Discriminant $D = b^2 - 4ac$: $D > 0$ two distinct real roots, $D = 0$ equal roots, $D < 0$ no real roots.
- **Arithmetic progression:** $a_n = a + (n - 1)d$. $S_n = (n/2)[2a + (n - 1)d] = (n/2)(a + l)$.
- **Triangles:** BPT (Thales): a line parallel to one side divides the other two sides in the same ratio. Similar triangles: AA, SSS, SAS. Ratio of areas = square of the ratio of corresponding sides.
- **Coordinate geometry:** Distance = $\sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$. Section formula: $((m\,x_2 + n\,x_1)/(m + n), (m\,y_2 + n\,y_1)/(m + n))$. Mid-point: $((x_1 + x_2)/2, (y_1 + y_2)/2)$.
- **Trigonometry:** $\sin\theta$ = opposite/hypotenuse, $\cos\theta$ = adjacent/hypotenuse, $\tan\theta$ = opposite/adjacent. $\sin^2\theta + \cos^2\theta = 1$. $1 + \tan^2\theta = \sec^2\theta$. $1 + \cot^2\theta = \operatorname{cosec}^2\theta$. $\sin 0°, 30°, 45°, 60°, 90° = 0, 1/2, 1/\sqrt{2}, \sqrt{3}/2, 1$. $\tan 30° = 1/\sqrt{3}$, $\tan 45° = 1$, $\tan 60° = \sqrt{3}$.
- **Heights and distances:** Angle of elevation (looking up) and angle of depression (looking down). Draw a right triangle and use $\tan\theta$ = height/distance.
- **Circles:** A tangent is perpendicular to the radius at the point of contact. Tangents from an external point are equal in length.
- **Areas related to circles:** Arc length = $(\theta/360) \times 2\pi r$. Area of sector = $(\theta/360) \times \pi r^2$.
- **Surface area and volume:** Combined solids: add volumes. Frustum: $V = (\pi h/3)(r_1^2 + r_2^2 + r_1 r_2)$, $CSA = \pi(r_1 + r_2)l$, $l = \sqrt{[h^2 + (r_1 - r_2)^2]}$.
- **Statistics:** Mean by direct, assumed mean or step-deviation method. Median = $l + [(n/2 - cf)/f] \times h$. Mode = $l + [(f_1 - f_0)/(2f_1 - f_0 - f_2)] \times h$. 3 Median = Mode + 2 Mean.
- **Probability:** $P(E)$ = favourable outcomes / total outcomes. $0 \le P(E) \le 1$. $P(E) + P(\text{not } E) = 1$. A pack has 52 cards; a die has 6 faces.

English

- **Tenses and agreement:** Use the correct tense consistently. Check subject-verb agreement in every sentence.
- **Reported speech:** Statement, question, command, request. Change tense, pronouns and time words.
- **Active and passive:** Convert correctly using the verb 'be' + past participle. The agent after 'by' is optional.
- **Determiners and modals:** Use few/a few, little/a little, some/any, much/many correctly. Modals show duty, advice, ability, possibility.
- **Gap filling and editing:** Read the whole sentence, then pick the word that fits grammar and meaning. In editing, correct only the marked errors.

- **Formal letter:** Sender's address, date, receiver, subject, salutation, body, 'Yours faithfully', name. Types: complaint, enquiry, order, application.
- **Analytical paragraph:** Study the chart or graph carefully. Write an introduction, main findings with figures, a comparison and a conclusion in the given word limit.
- **Reading and literature:** Answer comprehension in your own words. For literature questions, give a clear answer with reasons from the text.

Science

- **Chemical reactions:** Balance equations (same atoms on both sides). Types: combination, decomposition, displacement, double displacement, redox. Exothermic gives out heat. Rusting and rancidity are oxidation.
- **Acids, bases, salts:** pH scale 0 to 14: below 7 acidic, 7 neutral, above 7 basic. Common salt NaCl. Baking soda $NaHCO_3$. Washing soda $Na_2CO_3 \cdot 10H_2O$. Bleaching powder $CaOCl_2$. Plaster of Paris $CaSO_4 \cdot 1/2H_2O$.
- **Metals and non-metals:** Reactivity series: K, Na, Ca, Mg, Al, Zn, Fe, Pb, H, Cu, Hg, Ag, Au. Ionic compounds have high melting points. Prevent corrosion by galvanisation, painting, alloying.
- **Carbon compounds:** Covalent bonds. Allotropes: diamond, graphite, fullerene. Alkanes C_nH_{2n+2}, alkenes C_nH_{2n}, alkynes C_nH_{2n-2}. Ethanol and ethanoic acid. Soaps and detergents clean by forming micelles.
- **Life processes:** Autotrophic and heterotrophic nutrition. Photosynthesis. Human digestion. Aerobic and anaerobic respiration. Double circulation in the heart. Nephron filters blood. Xylem and phloem.
- **Control and coordination:** Neuron, reflex arc, brain. Hormones: insulin (pancreas), thyroxine (needs iodine), adrenaline, growth hormone. Plant hormones: auxin, gibberellin, cytokinin, abscisic acid, ethylene.
- **Reproduction:** Asexual: fission, budding, fragmentation, regeneration, spores. Sexual in plants: pollination and fertilisation. Human reproduction: male and female systems.
- **Heredity:** Mendel's monohybrid cross: phenotypic 3 : 1, genotypic 1 : 2 : 1. Dihybrid 9 : 3 : 3 : 1. Sex: XX female, XY male.
- **Light:** Mirror formula $1/v + 1/u = 1/f$. Magnification $m = h'/h = -v/u$. Snell's law: $n = \sin i / \sin r$. Lens formula $1/v - 1/u = 1/f$. Power $P = 1/f$ (dioptre).
- **Human eye:** Myopia (short sight) corrected by concave lens. Hypermetropia (long sight) corrected by convex lens. Dispersion by prism gives VIBGYOR. Scattering makes the sky blue.
- **Electricity:** $V = IR$. $R = \rho l/A$. Series: $R = R_1 + R_2$. Parallel: $1/R = 1/R_1 + 1/R_2$. $P = VI = I^2R$. $H = I^2Rt$. 1 kWh = 3.6×10^6 J.
- **Magnetic effects:** Oersted found that current makes a magnetic field. Right-hand thumb rule. Fleming's left-hand rule (motor), right-hand rule (generator). Domestic wiring: live (red), neutral (black), earth (green).
- **Our environment:** Food chain and web. Only about 10% of energy passes to the next trophic level. CFCs damage the ozone layer. Biodegradable and non-biodegradable waste.